

6) - Demuestre que:

$$a) \quad e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}$$

Prim $\rightarrow$ Sabemos que los elementos de una base son ortogonales a los elementos de una base dual, entonces no se que

$$e^i = \alpha (e_j \times e_k) \quad // \text{ como } e^i \cdot e_j = \delta^i_j$$

$$e_j \cdot e^i = e_j \cdot \alpha (e_j \times e_k)$$

$$1 = \alpha [e_j \cdot (e_j \times e_k)]$$

$$\alpha = \frac{1}{e_j \cdot (e_j \times e_k)}$$

$$\Rightarrow \boxed{e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}}$$

$$b) \quad N = e_1 \cdot (e_2 \times e_3) \wedge \bar{N} = e^1 \cdot (e^2 \times e^3) \Rightarrow N \bar{N} = 1$$

$$e' = \frac{e_2 \times e_3}{N} \Rightarrow e' = \frac{1}{N} \left(\overset{A}{\frac{e^3 \times e^1}{\bar{N}}} \right) \times \left(\overset{B}{\frac{e^1 \times e^2}{\bar{N}}} \right) \quad \begin{array}{l} \text{Propiedad triple} \\ \text{Producto} \\ \text{asociativo} \end{array}$$

$$= \frac{1}{N \bar{N}^2} \left(e^1 \left[\underbrace{e^2 \cdot (e^3 \times e^1)}_{\bar{N}} \right] - e^1 \left[\cancel{e^2 \cdot (e^1 \times e^1)} \right] \right)$$

$$= \frac{1}{N \bar{N}^2} (e^1 \bar{N})$$

$$\Rightarrow e' = \frac{1}{N \bar{N}} e^1 \Rightarrow \boxed{N \bar{N} = 1}$$

c) ¿qué vector satisface $a \cdot e^i = 1$? Demuestra que es único

Note que $a = a^i e_i$ y $e_i \cdot e^j = \delta_i^j$

$$\Rightarrow a^i e_i \cdot e^j = 1 \Rightarrow a^i \delta_i^j = 1 \Rightarrow \boxed{a^i e_i = e^j}$$

Supongamos que existen dos vectores a y b : $a \cdot e^i = 1$ y $b \cdot e^i = 1$

Note que

$$a \cdot e^i = b \cdot e^i$$

$$a \cdot e^i - b \cdot e^i = 0$$

$$(a-b) \cdot e^i = 0$$

$$(a-b) = \vec{0}$$

$$\boxed{a=b}$$

$$d) w_1 = 4\hat{i} + 2\hat{j} + \hat{k}, w_2 = 3\hat{i} + 3\hat{j}, w_3 = 2\hat{k}$$

$$\vec{a} \times \vec{b} = a^i w_i \times b^j w_j = a^i b^j \underbrace{(w_i \times w_j)}_{\propto e^k} = \propto a^i b^j e^k$$

I. Las bases recíprocas $\{e^i\}$

$$e^1 = \frac{w_2 \times w_3}{w_1 \cdot (w_2 \times w_3)} = \frac{(3\hat{i} + 3\hat{j}) \times 2\hat{k}}{(4\hat{i} + 2\hat{j} + \hat{k}) \cdot ((3\hat{i} + 3\hat{j}) \times 2\hat{k})} = \frac{6\hat{i} - 6\hat{j}}{(4\hat{i} + 2\hat{j} + \hat{k}) \cdot (6\hat{i} - 6\hat{j})}$$

$$e^1 = \frac{6\hat{i} - 6\hat{j}}{12} = \frac{1}{2}\hat{i} - \frac{1}{2}\hat{j}$$

$$e^2 = \frac{w_3 \times w_1}{w_2 \cdot (w_3 \times w_1)} = \frac{2\hat{k} \times (4\hat{i} + 2\hat{j} + \hat{k})}{(3\hat{i} + 3\hat{j}) \cdot (2\hat{k} \times (4\hat{i} + 2\hat{j} + \hat{k}))} = \frac{-4\hat{i} + 8\hat{j}}{12}$$

$$e^2 = -\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j}$$

$$e^3 = \frac{w_1 \times w_2}{w_3 \cdot (w_1 \times w_2)} = \frac{(4\hat{i} + 2\hat{j} + \hat{k}) \times (3\hat{i} + 3\hat{j})}{2\hat{k} \cdot ((4\hat{i} + 2\hat{j} + \hat{k}) \times (3\hat{i} + 3\hat{j}))} = \frac{12\hat{k} - 6\hat{i} + 3\hat{j} - 3\hat{i}}{12}$$

$$= \frac{-3\hat{i} + 3\hat{j} + 6\hat{k}}{12} = -\frac{1}{4}\hat{i} + \frac{1}{4}\hat{j} + \frac{1}{2}\hat{k}$$

II Covariantes a_i y contravariantes a^i

$$a \cdot e^i = a^j w_j \cdot e^i = a^j (w_j \cdot e^i)^{-1} = a^i \rightarrow \text{contravariantes}$$

$$a^1 = (1, 2, 3) \cdot (1/2, -1/2) = -1/2$$

$$a^2 = (1, 2, 3) \cdot (-1/3, 2/3) = 1$$

$$a^3 = (1, 2, 3) \cdot (-1/4, 1/4, 1/2) = 7/4$$

$$a \cdot w_i = a_i e^i \cdot w_i = a_i (e^i \cdot w_i) = a_i \rightarrow \text{covariantes}$$

$$a_1 = (1, 2, 3) \cdot (4, 2, 1) = 11$$

$$a_2 = (1, 2, 3) \cdot (3, 3) = 9$$

$$a_3 = (1, 2, 3) \cdot (0, 0, 2) = 6$$

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$$\langle \tilde{\sigma}_i | \sigma_i \rangle = \text{Tr}(\tilde{\sigma}_i^\dagger \sigma_i) = \delta_{ii}$$

$$\text{Tr}(\tilde{\sigma}_0^\dagger \sigma_0) = 1 \Rightarrow \text{Tr}(\tilde{\sigma}_0^\dagger) = 1 \Rightarrow \text{Tr} \begin{pmatrix} z_0^* & z_1^* \\ z_2^* & z_3^* \end{pmatrix} = 1$$

$$\text{Tr}(\tilde{\sigma}_1^\dagger \sigma_0) = 0 \Rightarrow \text{Tr}(\tilde{\sigma}_1^\dagger) = 0 \Rightarrow \text{Tr} \begin{pmatrix} z_1^* & z_2^* \\ z_3^* & z_4^* \end{pmatrix} = 0$$

$$\text{Tr}(\tilde{\sigma}_2^\dagger \sigma_0) = 0 \Rightarrow \text{Tr}(\tilde{\sigma}_2^\dagger) = 0 \Rightarrow \text{Tr} \begin{pmatrix} z_2^* & z_3^* \\ z_4^* & z_5^* \end{pmatrix} = 0$$

$$\text{Tr}(\tilde{\sigma}_3^\dagger \sigma_0) = 0 \Rightarrow \text{Tr}(\tilde{\sigma}_3^\dagger) = 0 \Rightarrow \text{Tr} \begin{pmatrix} z_3^* & z_4^* \\ z_5^* & z_6^* \end{pmatrix} = 0$$

$$\text{Tr}(\tilde{\sigma}_0^\dagger \sigma_1) = 0 \Rightarrow \text{Tr} \left[\begin{pmatrix} z_0^* & z_1^* \\ z_2^* & z_3^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \begin{pmatrix} z_1^* & z_0^* \\ z_3^* & z_2^* \end{pmatrix} = 0$$

$$\text{Tr}(\tilde{\sigma}_1^\dagger \sigma_1) = 1 \Rightarrow \text{Tr} \left[\begin{pmatrix} z_1^* & z_2^* \\ z_3^* & z_4^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \begin{pmatrix} z_2^* & z_1^* \\ z_4^* & z_3^* \end{pmatrix} = 1$$

$$\text{Tr}(\tilde{\sigma}_2^\dagger \sigma_1) = 0 \Rightarrow \text{Tr} \left[\begin{pmatrix} z_2^* & z_3^* \\ z_4^* & z_5^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \begin{pmatrix} z_3^* & z_2^* \\ z_5^* & z_4^* \end{pmatrix} = 0$$

$$\text{Tr}(\tilde{\sigma}_3^\dagger \sigma_1) = 0 \Rightarrow \text{Tr} \left[\begin{pmatrix} z_3^* & z_4^* \\ z_5^* & z_6^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \begin{pmatrix} z_4^* & z_3^* \\ z_6^* & z_5^* \end{pmatrix} = 0$$

$$T_1(\tilde{\sigma}_1^+ \sigma_1) = T_1 \left[\begin{pmatrix} z_1^0 & z_1^0 \\ z_1^0 & z_1^0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = T_1 \begin{pmatrix} z_1^0 & -iz_1^0 \\ iz_1^0 & -iz_1^0 \end{pmatrix} = 0$$

$$T_1(\tilde{\sigma}_1^+ \sigma_2) = T_1 \left[\begin{pmatrix} z_1^0 & z_1^0 \\ z_1^0 & z_1^0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] = T_1 \begin{pmatrix} iz_1^0 & -iz_1^0 \\ iz_1^0 & -iz_1^0 \end{pmatrix} = 0$$

$$T_1(\tilde{\sigma}_1^+ \sigma_3) = T_1 \left[\begin{pmatrix} z_1^0 & z_1^0 \\ z_1^0 & z_1^0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right] = T_1 \begin{pmatrix} iz_1^0 & -iz_1^0 \\ iz_1^0 & -iz_1^0 \end{pmatrix} = 1$$

$$T_1(\tilde{\sigma}_1^+ \sigma_4) = T_1 \left[\begin{pmatrix} z_1^0 & z_1^0 \\ z_1^0 & z_1^0 \end{pmatrix} \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix} \right] = T_1 \begin{pmatrix} iz_1^0 & -iz_1^0 \\ iz_1^0 & -iz_1^0 \end{pmatrix} = 0$$

$$T_2(\tilde{\sigma}_2^+ \sigma_1) = T_2 \left[\begin{pmatrix} z_2^0 & z_2^0 \\ z_2^0 & z_2^0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = T_2 \begin{pmatrix} z_2^0 & -z_2^0 \\ z_2^0 & -z_2^0 \end{pmatrix} = 0$$

$$T_2(\tilde{\sigma}_2^+ \sigma_2) = T_2 \left[\begin{pmatrix} z_2^0 & z_2^0 \\ z_2^0 & z_2^0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = T_2 \begin{pmatrix} z_2^0 & -z_2^0 \\ z_2^0 & -z_2^0 \end{pmatrix} = 0$$

$$T_2(\tilde{\sigma}_2^+ \sigma_3) = T_2 \left[\begin{pmatrix} z_2^0 & z_2^0 \\ z_2^0 & z_2^0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = T_2 \begin{pmatrix} z_2^0 & -z_2^0 \\ z_2^0 & -z_2^0 \end{pmatrix} = 0$$

$$T_2(\tilde{\sigma}_2^+ \sigma_4) = T_2 \left[\begin{pmatrix} z_2^0 & z_2^0 \\ z_2^0 & z_2^0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = T_2 \begin{pmatrix} z_2^0 & -z_2^0 \\ z_2^0 & -z_2^0 \end{pmatrix} = 1$$

$\Rightarrow$

Para $\tilde{\sigma}_0^+$

$$z_1^0 + z_4^0 = 1$$

$$\Rightarrow z_1^0 = z_4^0 \Rightarrow 2z_1^0 = 1 \Rightarrow z_1^0 = z_4^0 = 1/2$$

$$z_1^0 - z_4^0 = 0$$

$$z_2^0 + z_3^0 = 0$$

$$\Rightarrow z_2^0 = z_3^0 \Rightarrow 2z_2^0 = 0 \Rightarrow z_2^0 = z_3^0 = 0$$

$$i(z_2^0 - z_3^0) = 0$$

Para $\tilde{\sigma}_1^+$

$$z_1^1 + z_4^1 = 0$$

$$\Rightarrow z_1^1 = z_4^1 = 0$$

$$z_1^1 - z_4^1 = 0$$

$$z_2^1 + z_3^1 = 1$$

$$\Rightarrow z_2^1 = z_3^1 = 1/2$$

$$i(z_2^1 - z_3^1) = 0$$

Para $\tilde{\sigma}_2^+$

$$z_1^2 + z_4^2 = 0$$

$$\Rightarrow z_1^2 = z_4^2 = 0$$

$$z_1^2 - z_4^2 = 0$$

$$z_2^2 + z_3^2 = 0$$

$$\Rightarrow z_2^2 - z_3^2 = -i \Rightarrow 2z_2^2 = -i \quad z_2^2 = -i/2 \wedge z_3^2 = i/2$$

$$i(z_2^2 - z_3^2) = 1$$

$$z_2^2 = -z_3^2$$

Para $\tilde{\sigma}^{\pm}$

$$\tilde{z}_1^{\pm} + \tilde{z}_3^{\pm} = 0$$

$$\Rightarrow \tilde{z}_1^{\pm} = -\tilde{z}_3^{\pm} \Rightarrow 2\tilde{z}_1^{\pm} = 1 \Rightarrow \tilde{z}_1^{\pm} = 1/2 \wedge \tilde{z}_3^{\pm} = -1/2$$

$$\tilde{z}_1^{\pm} - \tilde{z}_3^{\pm} = 1$$

$$\tilde{z}_1^{\pm} + \tilde{z}_3^{\pm} = 0$$

$$i(\tilde{z}_1^{\pm} - \tilde{z}_3^{\pm}) = 0$$

$$\tilde{\sigma}^{0+} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} = \tilde{\sigma}_0 \quad ; \quad \tilde{\sigma}^{1+} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} = \tilde{\sigma}_1$$

$$\tilde{\sigma}^{2+} = \begin{pmatrix} 0 & -i/2 \\ i/2 & 0 \end{pmatrix} = \tilde{\sigma}_2 \quad ; \quad \tilde{\sigma}^{3+} = \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = \tilde{\sigma}_3$$

Note que para una matriz cualquiera $a \in M_2$ su 1-forma asociada será